Distributed Systems
Exercise Sheet 4, Thursday, 14:15
Klingemann, SS 2026

Deadline: 28th May 2026

1st Assessed Exercise

1. Communication via Sockets
Understand the example code and test the programs. Client and server can run on the same machine.
Hint: You can address your own machine using the name `localhost`.

2. Realize a remote method invocation by using plain sockets (without any middleware)
Use TCP-sockets for this task.
Extend the client and server so that the client can invoke methods on the server and a corresponding return value is delivered to the client. Therefore, you have to extend your simple system for the management of equity funds and their stocks from Sheet 3. Use <u>two</u> (equity) Fund-objects on the server. Create a class FundProxy that is used only within the client. This class is providing methods with the same signatures as the class Fund. For each object of the class Fund on the server, there exists a corresponding object of the class FundProxy in the client. Objects of the class Fund only exist on the server and must not be used on the client.
The client should be able to invoke the four methods defined in Sheet 3 on the Fund-objects. The client is doing this by invoking the method on the corresponding proxy-object. The proxy-object is forwarding the request to the server. On the server the method is invoked on the corresponding object of the class Fund and a return value is sent back to the proxy-object.
Hence, to solve this task, you have to transport the information about the object, the method and the corresponding parameters by means of sockets. Similarly, you have to transport the return-value. Solve this task by creating a class Message. A message-object should contain different attributes for the different pieces of information that are necessary for the method invocation. This object can then be serialized using the class `ObjectOutputStream`. The serialized object is then transported using TCP-sockets. The return-value is also transported using TCP-sockets but need not be a message-object.

Organisational matters

- You have to solve the exercise completely on your own! (No working in groups!)
- You have to write the code yourself and must not generate it by AI-tools etc.
- It is necessary but not sufficient to present a working program. Moreover, you have to be able to explain all parts of your program, be able to answer questions with respect to your program and make small extensions of your program.
- <u>Your program has to be created completely within the exercise slot.</u>
- If you violate one of the rules above, this implies that you definitely fail in this exercise.
- You can only present solutions that correspond to the exercise slot you are assigned to.
- It is in your responsibility to present your solution in time before the deadline. The assessment of your solution can only be guaranteed if you finish your program 60 minutes before the end of the exercises.
- To take part in the exam it is required to solve at least three of five assessed exercise sheets.